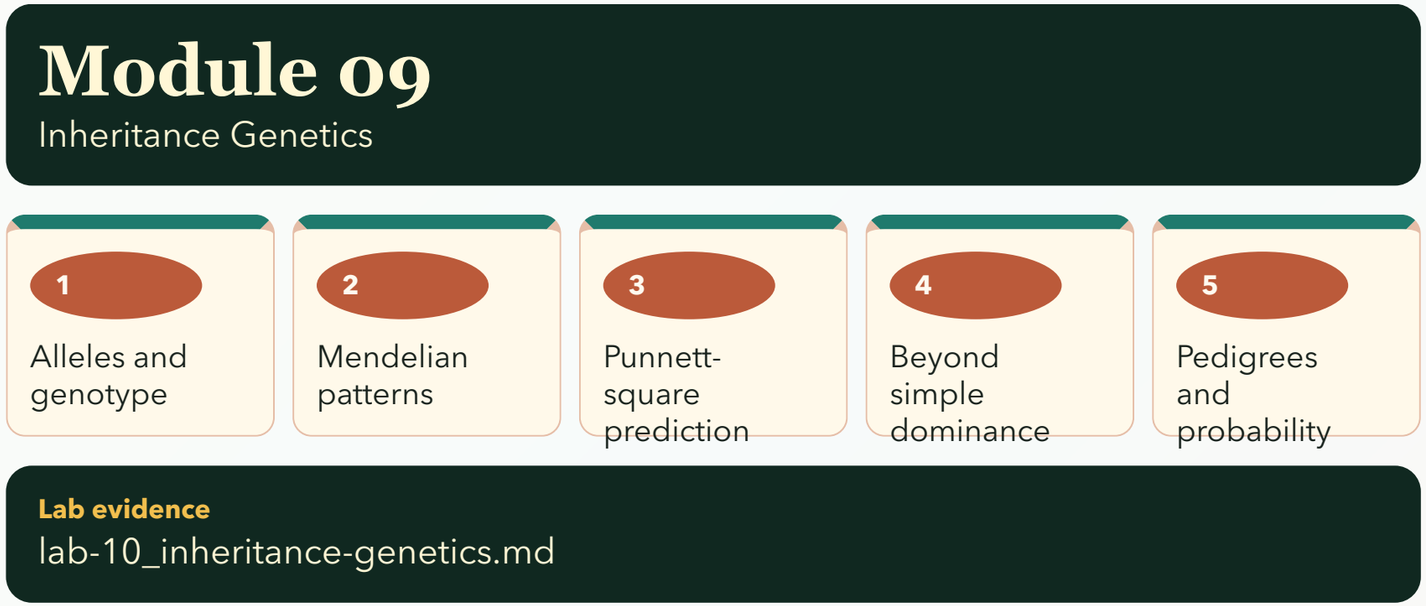
INHERITANCE GENETICS

Module map

- Module 09: Inheritance Genetics
- Connected lab: lab-10_inheritance-genetics.md
- Use this deck as the visual path through the module's evidence and retrieval work.



INHERITANCE GENETICS

Learning objectives

- Distinguish gene, allele, genotype, and phenotype.
- Use segregation to explain monohybrid inheritance.
- Build and interpret Punnett-square probabilities.
- Recognize inheritance patterns beyond complete dominance.
- Use pedigree evidence to reason about traits.

OBJECTIVE 1
Distinguish gene, allele, genotype, and phenotype.

OBJECTIVE 2
Use segregation to explain monohybrid inheritance.

OBJECTIVE 3
Build and interpret Punnett-square probabilities.

OBJECTIVE 4
Recognize inheritance patterns beyond complete dominance.

OBJECTIVE 5
Use pedigree evidence to reason about traits.

INHERITANCE GENETICS

Topic sequence

- Alleles and genotype: Alleles are gene variants that contribute to genotype and phenotype.
- Mendelian patterns: Mendelian inheritance predicts patterns when alleles segregate into gametes.
- Punnett-square prediction: Punnett squares model probability, not guaranteed outcomes.
- Beyond simple dominance: Incomplete dominance, codominance, polygenic traits, and environment complicate simple patterns.
- Pedigrees and probability: Pedigrees use family evidence to infer inheritance patterns.

01

Alleles and genotype

Alleles are gene variants that contribute to genotype and phenotype.

02

Mendelian patterns

Mendelian inheritance predicts patterns when alleles segregate into gametes.

03

Punnett-square prediction

Punnett squares model probability, not guaranteed outcomes.

04

Beyond simple dominance

Incomplete dominance, codominance, polygenic traits, and environment complicate simple patterns.

05

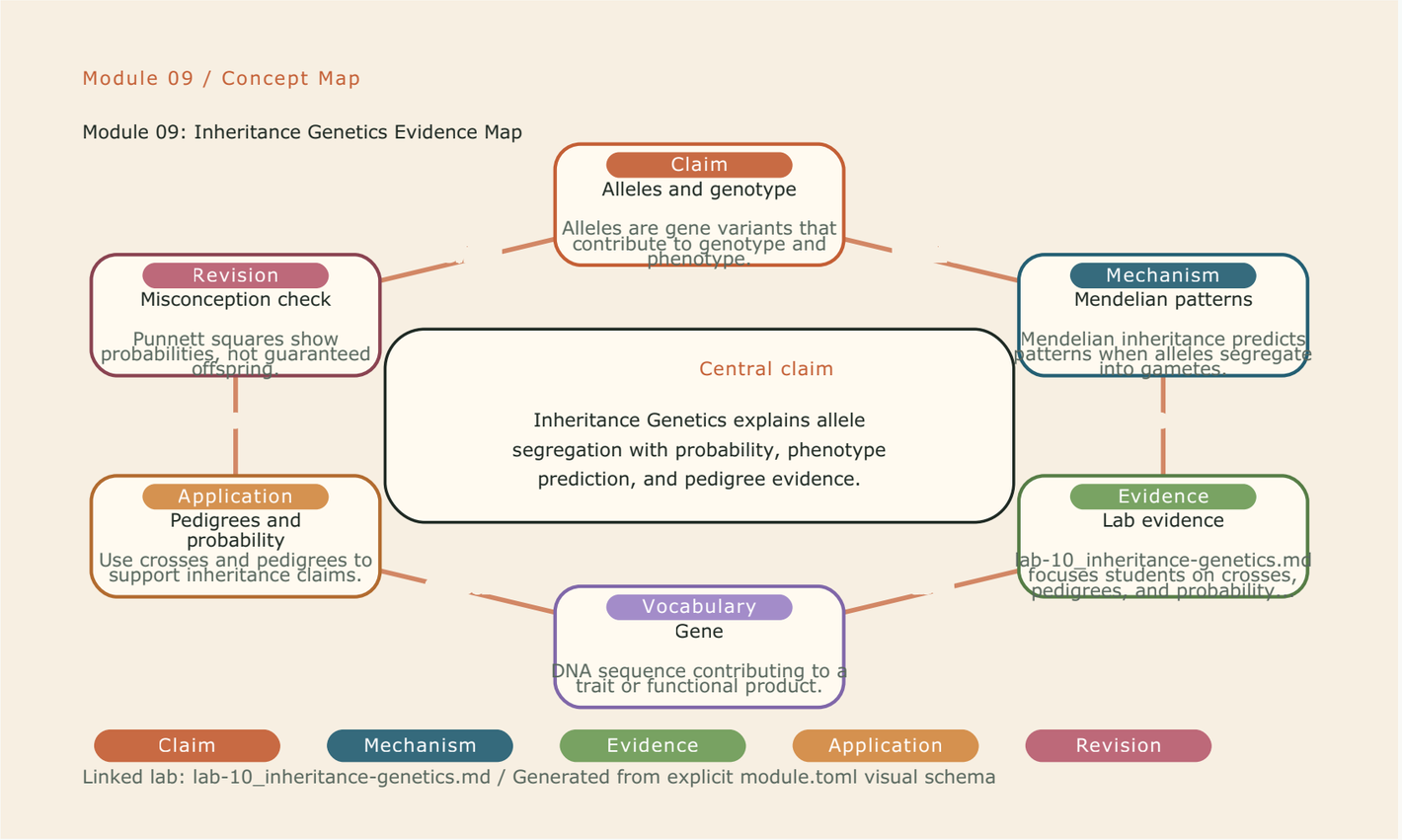
Pedigrees and probability

Pedigrees use family evidence to infer inheritance patterns.

INHERITANCE GENETICS

Module 09: Inheritance Genetics Evidence Map

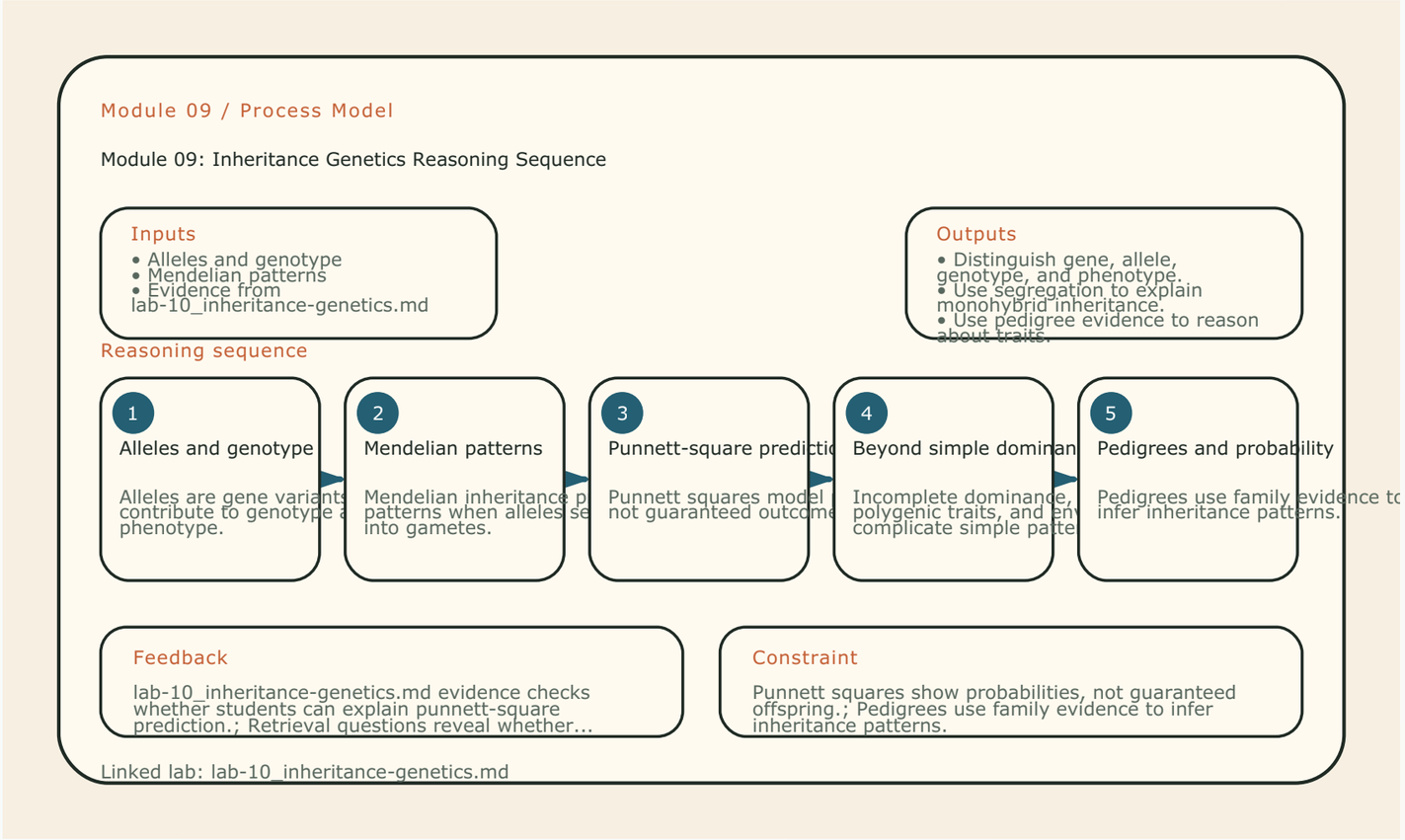
- Module focus: Inheritance Genetics
- Central claim: Inheritance Genetics explains allele segregation with probability, phenotype prediction, and pedigree evidence.
- Key nodes to connect: Alleles and genotype, Mendelian patterns, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-10_inheritance-genetics.md supplies an evidence surface for the map.



INHERITANCE GENETICS

Module 09: Inheritance Genetics Reasoning Sequence

- Module focus: Inheritance Genetics
- Trace the model: Alleles and genotype -> Mendelian patterns -> Punnett-square prediction -> Beyond simple dominance
- Inputs become outputs: Alleles and genotype, Mendelian patterns -> Distinguish gene, allele, genotype, and phenotype., Use segregation to explain monohybrid inheritance.
- Feedback check: lab-10_inheritance-genetics.md evidence checks whether students can explain punnett-square prediction.
- Lab connection: lab-10_inheritance-genetics.md gives students a process to observe or test.



INHERITANCE GENETICS

Terms as evidence handles

- Gene: DNA sequence contributing to a trait or functional product.
- Allele: Variant form of a gene.
- Genotype: Allele combination of an organism.
- Phenotype: Observable trait or characteristic.
- Dominant: Allele effect visible with one copy in simple dominance.
- Recessive: Allele effect masked by dominant allele in heterozygotes.

GENE
DNA sequence contributing to a trait or functional product.

ALLELE
Variant form of a gene.

GENOTYPE
Allele combination of an organism.

PHENOTYPE
Observable trait or characteristic.

DOMINANT
Allele effect visible with one copy in simple dominance.

RECESSIVE
Allele effect masked by dominant allele in heterozygotes.

INHERITANCE GENETICS

Lab connection

- Lab file: lab-10_inheritance-genetics.md
- Lab evidence should test or illustrate: Punnett-square prediction
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Alleles and genotype

EVIDENCE
lab-10_inheritance-genetics.md

CLAIM
Distinguish gene, allele, genotype, and phenotype.

INHERITANCE GENETICS

Contrast check

- Do not stop at: Alleles are gene variants that contribute to genotype and phenotype.
- Stronger explanation: Pedigrees use family evidence to infer inheritance patterns.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Alleles are gene variants that contribute to genotype and phenotype.

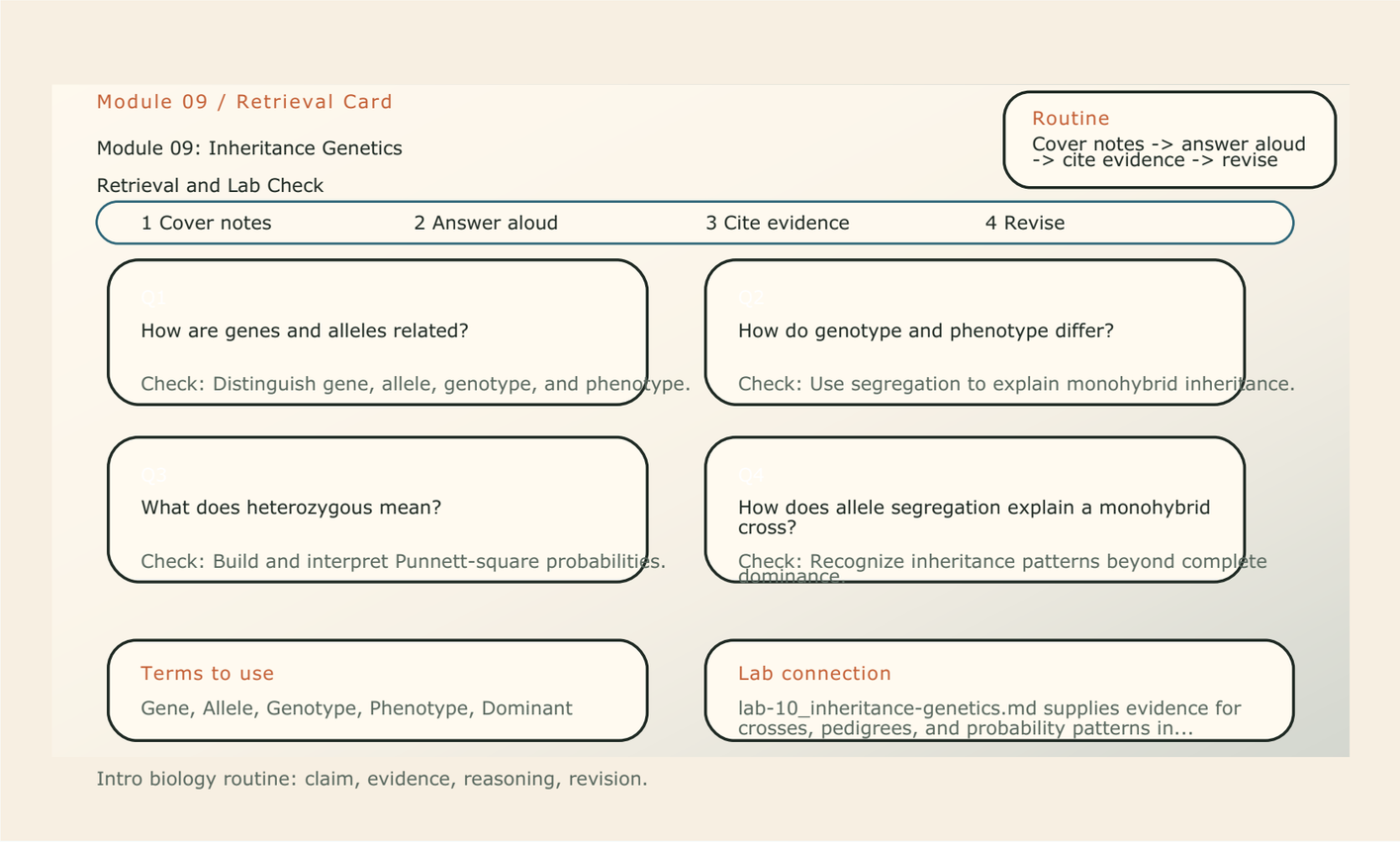
DEEPER

Pedigrees use family evidence to infer inheritance patterns.

INHERITANCE GENETICS

Module 09: Inheritance Genetics Retrieval and Lab Check

- Module focus: Inheritance Genetics
- Retrieval prompt: How are genes and alleles related?
- Answer check: Distinguish gene, allele, genotype, and phenotype.
- Required terms: Gene, Allele, Genotype
- Lab connection: lab-10_inheritance-genetics.md; lab-10_inheritance-genetics.md supplies evidence for crosses, pedigrees, and probability patterns in inheritance.



INHERITANCE GENETICS

Practice quiz bridge

- 1. An organism's genotype is its:
- 2. Two heterozygotes are crossed (Aa x Aa). What fraction of offspring are expected to show the recessive trait?
- 3. A heterozygote (Aa) shows the dominant trait because:
- 4. What is a pedigree used for?

QUESTION 1

Answer first

An organism's genotype is its:

QUESTION 2

Answer first

Two heterozygotes are crossed (Aa x Aa). What fraction of offspring are expected to show the recessive trait?

QUESTION 3

Answer first

A heterozygote (Aa) shows the dominant trait because:

QUESTION 4

Answer first

What is a pedigree used for?

INHERITANCE GENETICS

Synthesis and exit ticket

- Exit claim: explain alleles and genotype using evidence from lab-10_inheritance-genetics.md.
- Must include: Gene, Allele, Genotype.
- Revision prompt: what evidence would change your explanation?

CLAIM
Alleles and genotype

EVIDENCE
lab-10_inheritance-genetics.md

REVISION
What would change your mind?